FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 1

PHYSICS / APPLIED PHYSICS

MATERIALS SCIENCE

Module I & II: Materials, Interatomic Bonding, and Crystalline Solids

Maximum Time: 2 Hours		**Maximum Marks: 70**
Section A		
All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)		
1	Distinguish between the subdisciplines of "materials science" and "materials engineering". State the four components of the central paradigm of materials science and engineering.	**3**
2	What are the three primary criteria that engineers and scientists must consider when selecting a material for a specific engineering application?	**3**
3	What are composites? How do metals differ from ceramics in terms of atomic bonding and mechanical properties?	**3**
4	Define smart materials. What are the two essential components of a smart material system, and what functions do they perform?	**3**
5	Define bonding energy E_0 between two atoms in a solid. How do the magnitude of E_0 and the shape of the potential energy curve influence the melting temperature and thermal expansion of a material?	**3**
6	Explain metallic bonding in crystals using the electron cloud model. Why are metals good conductors of electricity and heat based on their bonding nature?	**3**
7	What are van der Waals forces? Explain what a permanent dipole bond is and how it affects intermolecular interactions.	**3**
8	Why does solid ice have a lower density than liquid water at room temperature? State two familiar natural or practical consequences of this anomalous volume expansion upon freezing.	**3**
9	Explain the terms single crystal, polycrystalline material, and grain boundary in crystallography.	**3**
10	Define anisotropy in crystalline materials. How is the directionality of properties related to atomic spacing and crystal symmetry?	**3**
Section B		
All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)		
11	Discuss the modern materials' needs in solar energy conversion, transportation, hydrogen fuel cells, and environmental quality. Why is it important to consider the "cradle-to-grave" life cycle of materials?	**6**
12	Explain how the optical transmittance properties (transparency, translucency, and opacity) of aluminum oxide (Al_2O_3) disk specimens depend on their internal structure and processing. What is a semiconducting material, and what is its significance?	**6**
13	Explain the concept of bonding forces and energies in solids. The potential energy between two atoms in a solid is given by $U(r) = -\frac{A}{r} + \frac{B}{r^n}$, where A and B are constants, and r is the interatomic distance. Explain what happens to the energy when r increases or decreases, and derive the expression for the equilibrium bond length r_0.	**6**
14	The atomic radii of K^+ and Br^- ions are $0.138\ nm$ and $0.196\ nm$, respectively. (a) Using the coulombic attractive energy relation $E_A = -\frac{1}{4\pi\varepsilon_0 r}(\lvert Z_1\rvert e)(\lvert Z_2\rvert e)$, calculate the force of attraction between these two ions at their equilibrium interionic separation (when the ions just touch one another). (b) What is the force of repulsion at this same separation distance? Take $\varepsilon_0 = 8.85 \times 10^{-12}\ F/m$ and $e = 1.602 \times 10^{-19}\ C$.	**6**
15	Explain mixed covalent–ionic bonding using the bonding tetrahedron. Compute the percent ionic character (\%IC) and percent covalent character of the interatomic bond that forms between carbon and hydrogen, given that the electronegativities for C and H are $X_C = 2.5$ and $X_H = 2.1$, respectively.	**6**
16	Water expands by approximately 9 volume percent upon freezing, unlike most substances that contract. Explain in detail the molecular structure of H_2O, nearest-neighbor coordination, and the hydrogen bonding interactions in solid ice versus liquid water responsible for this behaviour.	**6**
17	Describe the various stages in the solidification of a polycrystalline specimen with neat schematic diagrams. Why are polycrystalline materials typically isotropic while single crystals	**6**

	exhibit anisotropic behaviour, and how is "magnetic texture" utilized to minimize energy losses in transformer cores?	
18	What are nanocrystalline solids (nanomaterials)? Distinguish between the "top-down" and "bottom-up" approaches in materials science, and explain why the physical and chemical characteristics of materials undergo dramatic changes as particle size approaches nanoscale dimensions.	**6**
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	Classify solid materials into metals, ceramics, polymers, and composites based on their chemical makeup and atomic structure. Compare their density, mechanical (stiffness, strength, fracture toughness), electrical, and optical properties with suitable examples, and discuss the four classes of advanced materials used in high-technology applications.	**10**
20	Discuss the primary (ionic, covalent, and metallic) and secondary (fluctuating induced dipole, polar molecule–induced dipole, and permanent dipole/hydrogen) interatomic bonds in solids. Explain sp^3 and sp^2 bond hybridization in carbon, and analyze how the nature of bonding in diamond and graphite contributes to their hardness, electrical conductivity, and thermal properties.	**10**